Team Task Tracker — Setup & Running Manual

Official Guide for Cloning, Installing Dependencies, and Running the Application on Any Machine

1. System Prerequisites

Ensure the target computer has the following tools installed:

Component	Minimum Version	Purpose
Python	3.10 or higher	Django Backend REST API & ORM
Node.js	18.x or higher (with npm)	Next.js 16 / React Frontend UI
MongoDB	6.0+ or MongoDB Atlas	Primary Database Storage (Port 27017)

2. Clone or Extract the Project

```
git clone https://github.com/SAfsehEhsani/Team-Task-Tracker-Jira.git
cd Team-Task-Tracker-Jira
```

3. Method 1: 1-Click Automated Startup (Recommended for Windows)

In the root project folder, simply double-click **start.bat** or execute via Command Prompt:

```
.\start.bat
```

What this script does automatically:

- Verifies that the local MongoDB database service is running.
- Detects missing backend_venv, creates a fresh virtual environment specifically for your machine, and installs all packages from backend\requirements.txt.
- Detects missing frontend\node_modules and automatically runs npm install.
- Starts Django Backend on **http://127.0.0.1:8000** and Next.js Frontend on **http://localhost:3000**.
- Automatically launches your default web browser to the application dashboard.

4. Method 2: Manual Step-by-Step Commands

Step A: Ensure MongoDB Service is Active

```
net start MongoDB
```

Step B: Terminal 1 — Backend (Django REST Framework)

```
:: 1. Create virtual environment & activate (first time)
python -m venv backend_venv && backend_venv\Scripts\activate
:: 2. Install dependencies & launch server
pip install -r backend\requirements.txt
python backend\manage.py runserver 127.0.0.1:8000
```

Step C: Terminal 2 — Frontend (Next.js)

```
cd frontend
npm install
npm run dev
```

5. macOS & Linux Users

```
chmod +x start.sh && ./start.sh
```

6. Accessing the Application & Demo Credentials

- **Frontend Web UI:** http://localhost:3000 | **Backend REST API:** http://127.0.0.1:8000/api/health/

Role	Email Address	Password	Permissions & Access
Admin	admin@teamtracker.local	Admin@12345	Manage teams, invite members, assign tasks, live MongoDB data inspector
Member	alice.chen@techcorp.io	Alice@12345	View assigned tasks, update task workflow status, view team timeline

*Note: Users can also click **'Create an account'** on the login page to register custom Administrator or Member accounts.*

7. Technical Note: Why backend_venv & node_modules are Excluded from Git

Standard software engineering protocol dictates that virtual environments and node_modules are strictly excluded via .gitignore. Virtual environments hardcode creator-specific absolute machine paths (e.g. c:\users\syed afseh ehsani\...) and machine-specific compiled C-binaries (e.g. Windows .pyd wheels). Committing them causes fatal crashes on other computers. Instead, backend/requirements.txt and frontend/package.json allow any clone to construct an error-free native environment automatically in seconds using start.bat.